LAB MANAGEMENT SYSTEM

UI MOCKUP & FUNCTIONALITY BLUEPRINT

Platforms React.js web • Flutter mobile

Backend Express.js **Database** MySQL

Themes Dark and light

Purpose

A school laboratory resource management system for tracking equipment, lending and returns, resource condition, user accounts, notifications, approvals and audit history.

DESIGN PRINCIPLE	WHAT IT MEANS
Accountability	Every lending, return, inspection and correction request is tied to a user and timestamp.
Condition before lending	A lab technician must inspect the resource before it can be issued.
Protected records	Return records cannot be silently overwritten; corrections require borrower and authorized administrator approval.
Least privilege	Each role sees only the screens and actions permitted for that role.

This document is a proposed mockup and requirements blueprint. Sample figures and identifiers are illustrative, not live school data.

1. Product overview

The system has a management workspace for laboratory staff and a simpler self-service workspace for students and school staff. Both platforms share the same business rules and permissions.

PLATFORM	PRIMARY USE	MAIN UI PATTERN
React.js web	Lab administration and detailed record management	Sidebar, top bar, data tables, filters, forms and reports
Flutter mobile	Borrower self-service, confirmations and notifications	Bottom navigation, cards, compact lists, touch-friendly actions
Express.js API	Authentication, permissions, validation and transaction workflows	REST endpoints with server-side authorization
MySQL	Users, resources, transactions, approvals, notifications and audit records	Relational data and transaction-safe updates

Brand and visual language

ELEMENT	PROPOSED DESIGN
Product name	Lab Management System (working name; LabCore may be used as a brand label).
Primary color	Teal #0D9488 for primary actions and active navigation.
Dark theme	Navy #172033 backgrounds, slate panels, white text and teal highlights.
Light theme	White surfaces, pale gray #F1F5F9 backgrounds, navy text and teal highlights.
Status colors	Green = available/approved; amber = pending/attention; red = damaged/overdue/blocked.
Accessibility	Readable contrast, clear labels, visible focus states, and icons paired with text.

2. User roles and permissions

Account creation is performed by an authorized administrator or lab assistant with account-management permission. Any approval permission must be assigned explicitly.

ROLE	KEY PERMISSIONS
Lab Assistant / Lab Technician	Manage inventory; inspect resources; issue and receive materials; record condition and responsibility marks; manage correction requests and view permitted reports.
Student	View own borrowed items, due dates, return records and notifications; confirm or dispute correction requests affecting their records.
Teacher / School Staff	View own account and borrowing history; borrow eligible items; receive notifications and respond to relevant confirmations.
Head of Department (HOD)	View authorized departmental reports and relevant laboratory activity; approve changes only if explicitly granted approval permission.
Director of Studies (DOS)	View authorized class/program summaries and relevant reports; receive notifications; no automatic access to all private records.
Director of Discipline (DOD)	View permitted incidents or overdue summaries; receive notifications; no unrestricted inventory editing by default.
School Administrator	Create/manage accounts and roles; oversee inventory, reports and audit history; provide the second correction approval if authorized.

Permission rules

- Authorization must be checked by the Express.js backend, not only by hiding buttons in the UI.
- Users see their own borrowing history unless a role explicitly grants broader access.
- Sensitive actions such as role changes, resource write-offs and corrections must be logged.

3. Navigation and screen inventory

SCREEN	WEB EXPERIENCE	MOBILE EXPERIENCE
Login & account access	Login, password reset, session expiry, access denied	Compact login and password reset
Dashboard	Role-based metrics, activity, alerts and quick actions	Personal summary cards and alerts
Inventory	Search, filters, categories, locations, status and item detail	Searchable list; management actions restricted by role
User management	Create accounts, assign role, class/program/level, activate/deactivate	Authorized account managers only
Issue / lending	Select borrower/resource, inspect condition, confirm issue	Quick issue flow for authorized lab staff
Returns	Find active loan, inspect return condition, record time and outcome	Borrower reviews and confirms own return details
Correction requests	Compare original/proposed values, approve/reject and record reason	Review and respond to pending requests
Notifications	Filters, read/unread and action links	Notification list and detail
Reports & audit	Inventory, usage, overdue, damage and change history	Summary views; exports restricted by permission
Settings & profile	Profile, password, theme and notification preferences	Profile, password, theme and notification preferences

Suggested web sidebar

Dashboard → Inventory → Users → Lending → Returns → Correction Requests → Notifications → Reports → Audit History → Settings

Suggested mobile navigation

Home → My Materials (or Inventory for lab staff) → Notifications → Profile. Role-specific shortcuts appear on the home screen.

4. Web dashboard mockup

Illustrative desktop layout. Counts below are placeholders and must come from MySQL in the real application.

LAB MANAGEMENT SYSTEM Laboratory workspace	Search resources, users… Notifications • Profile

SIDEBAR

Dashboard
Inventory
Users
Lending
Returns
Correction requests
Notifications
Reports
Audit history
Settings

Laboratory overview
Monitor resources, lending and returns.

120 Total computers	**86** Available items
28 Items borrowed	**6** Overdue items

Quick actions
[+ Add resource] [Issue material] [Receive return] [Correction requests]

Recent activity
• PC 043/NYH — condition checked
• Keyboard KB-018 — returned
• Return correction — awaiting borrower approval

Dashboard cards change by role. Students and teachers see personal loans, deadlines and notifications rather than management metrics.

5. Inventory and resource condition UI

Each resource has a unique identifier, category, location, condition/status and history. Computers can use a display identifier such as 043/NYH.

FIELD	EXAMPLE / RULE
Resource ID	Internal ID such as PC-043-NYH; display label 043/NYH.
Category	Computer, monitor, keyboard, mouse, cable, adapter, projector or other.
Location	Laboratory, building, room or shelf reference.
Condition	Excellent, Good, Damaged, Faulty.
Operational status	Available, Borrowed, Under maintenance, Retired/unavailable.
Accessories	Power cable, mouse, keyboard or other included components.
Inspection evidence	Notes and optional photos if enabled by school policy.
History	Inspections, lending, returns, maintenance and approved changes.

Mandatory condition check before lending

STEP	UI ACTION	SYSTEM RULE
1	Select resource and borrower	Resource must exist and be eligible for lending.
2	Inspect function and physical condition	Technician records the condition before issue.
3	Check accessories and existing damage	Record missing parts or damage in inspection notes.
4	Save pre-lending inspection	Inspection is timestamped and linked to the technician.
5	Confirm issue	Block faulty, under-maintenance, retired, already-borrowed or unavailable resources.

Condition describes the physical/functional state. Operational status describes whether an item may be lent. These should be separate fields.

6. Lending and return transaction mockups

Lending form — suggested fields

FIELD / CONTROL	EXPECTED BEHAVIOUR
Borrower search	Search by user ID, name, email, class or program/level.
Resource search	Search by identifier, category, location or barcode/QR if later enabled.
Pre-lending condition	Required inspection and notes; show last known condition.
Accessories included	Checkbox list appropriate to the resource.
Expected return time	Required if the school uses due dates.
Borrower acknowledgement	Confirm the item and recorded condition were reviewed.
Issue confirmation	Create a traceable loan event and set resource status to Borrowed.

Return form — suggested fields

FIELD / CONTROL	EXPECTED BEHAVIOUR
Active loan	Show borrower, resource and issue details.
Actual return time	Use server timestamp; corrections go through the request workflow.
Return condition	Required inspection after return: Excellent, Good, Damaged or Faulty.
Accessories returned	Record whether all issued components came back.
Condition comparison	Compare issue vs return condition and flag possible new damage.
Responsibility marks	Use school-defined rubric; store mark, reason and recorder.
Borrower confirmation	Borrower can acknowledge the return or raise a dispute.
Final status	Available only if eligible; otherwise Under maintenance or Unavailable.

The school must define the mark scale and criteria. A damaged item alone does not prove that the borrower caused the damage.

7. Protected records and correction approvals

A completed return record must not be silently edited. A correction request preserves the original record and stores proposed values, reason and approval history.

STAGE	USER ACTION	SYSTEM BEHAVIOUR
1. Request	Technician proposes correction and provides a reason.	Create pending request; original return remains unchanged.
2. Borrower review	Affected borrower reviews old and proposed values.	Record approval or dispute with identity and timestamp.
3. Administrator review	Authorized administrator reviews the request.	Record approval or rejection, identity, timestamp and reason.
4. Apply approved change	System applies the approved correction.	Preserve original and corrected values in audit history.
5. Notify	Affected parties receive a notification.	Notify borrower, technician and authorized approver.

Both the affected borrower and an authorized administrator must approve a correction before it is applied. If either rejects it, the correction is not applied; a school-defined dispute process should handle disagreements.

Security requirements

• Enforce approval rules in Express.js; a modified frontend must not bypass them.

• Store original values, proposed values, reason, requester, approvers, decisions and timestamps.

• Normal UI workflows must not delete audit records.

• Protect audit records with restricted database permissions, controlled admin access and backups.

8. Notifications, reports and audit history

EVENT	RECIPIENTS / ACTION
Material issued	Borrower gets item details, recorded condition and due date if applicable.
Due date approaching / overdue	Borrower and permitted lab staff get reminders.
Material returned	Borrower gets return summary and recorded condition.
Correction requested	Affected borrower and authorized administrator are notified.
Correction approved/rejected	Requester and affected parties receive the outcome.
Resource faulty / under maintenance	Lab staff receive an operational alert.
Low stock / missing accessory	Authorized staff receive an alert if configured.

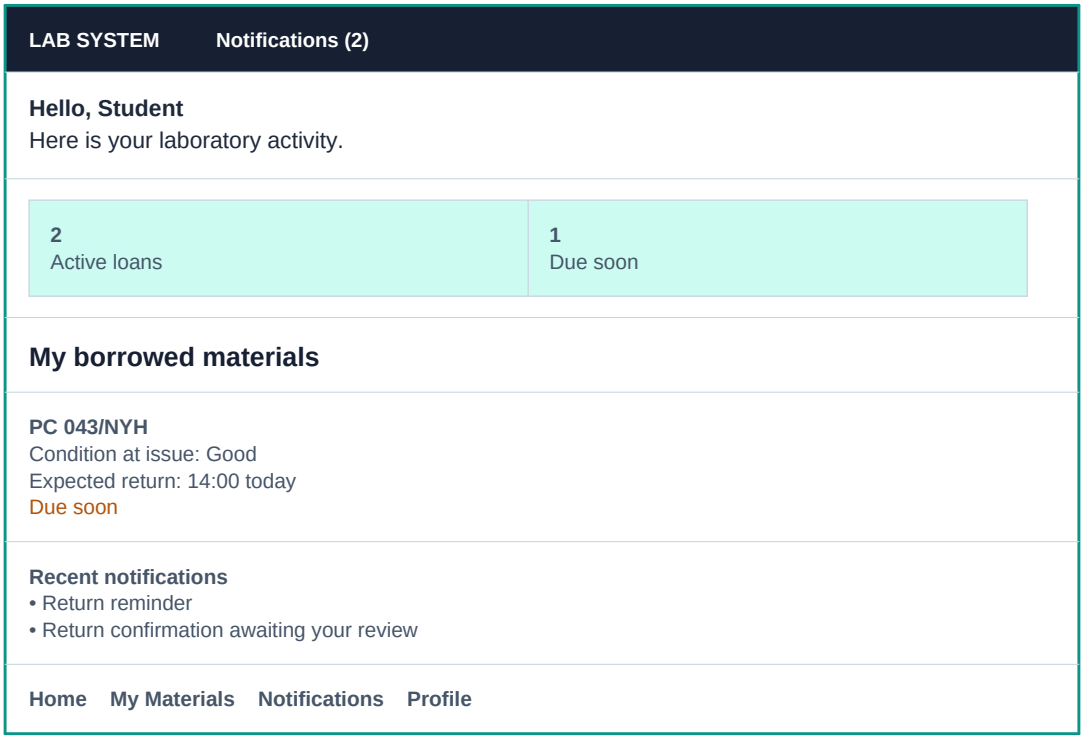
Reports

REPORT	USE
Inventory register	Resources by category, location, condition and status.
Borrowing/return history	Borrower, resource, issue/return timestamps and transaction status.
Overdue materials	Open loans past due date, where due dates are enabled.
Condition and maintenance	Damaged/faulty items, maintenance periods and inspection history.
User/class/level summary	Authorized view of borrowing activity by class or program/level.
Responsibility marks	Marks with rubric, reason and recorder; access limited by role.
Correction and audit report	Requests, decisions, original/corrected values and timestamps.

PDF/CSV exports should be limited to authorized roles and avoid exposing unnecessary personal information.

9. Flutter mobile mockup

Borrower home screen — conceptual layout



Lab assistant mobile experience

• Dashboard with pending returns, unavailable items and correction requests.

• Search resource or borrower; inspect resource before issue.

• Record return condition, accessories and borrower acknowledgement.

• Review and respond to correction requests only within granted permissions.

10. Core data model (planning level)

This is a logical starting point, not final SQL. Finalize the schema before backend implementation.

ENTITY	IMPORTANT DATA
users	id, user_code, full_name, email/phone, password_hash, role_id, class/program/level, account_status
roles / permissions	Role definitions and explicit permission mapping.
resources	id, asset_code, name, category_id, location_id, condition, operational_status, serial_number
categories / locations	Reusable resource categories and laboratory/room/location records.
resource_inspections	resource_id, inspector_id, type, condition, notes, accessories, created_at
loans	resource_id, borrower_id, issued_by, issued_at, due_at, status, issue_inspection_id
returns	loan_id, received_by, returned_at, return_inspection_id, acknowledgement, responsibility_mark, mark_reason
correction_requests	target_record, requester_id, reason, original_snapshot, proposed_values, status
correction_approvals	request_id, approver_id, approval_type, decision, comment, decided_at
notifications	recipient_id, type, message, related_record, read_at, created_at
audit_logs	actor_id, action, entity_type, entity_id, before/after snapshot, reason, timestamp
maintenance_records	resource_id, reported_by, issue, status, opened_at, closed_at, resolution

Recommended implementation controls

- Hash passwords; validate inputs; use secure token/session handling and HTTPS in deployment.
- Use database transactions for issue/return operations so loan and resource status stay consistent.
- Index resource codes, borrower IDs, loan status, due dates and notification recipients.
- Use server timestamps and consistent timezone handling.
- Create database backup and restore procedures.

11. Functional acceptance checklist

ID	REQUIREMENT	DONE WHEN
F-01	Role-based access	Each role sees permitted screens/actions; backend rejects unauthorized requests.
F-02	Account creation	Authorized staff create accounts and classify students by REB class or RTB level/program.
F-03	Inventory	Resources have unique IDs, categories, locations, condition and operational status.
F-04	Pre-lending inspection	Technician saves condition/accessory inspection before issue.
F-05	Block unsafe issue	Faulty, under-maintenance, retired, borrowed or unavailable items cannot be lent.
F-06	Lending	Issue records borrower, resource, technician, time, condition and due date if used.
F-07	Return inspection	Return records condition, accessories, time and comparison with issue condition.
F-08	Responsibility marks	School-defined rubric, mark and reason are stored with the recorder.
F-09	Borrower review	Borrowers view their own transactions and acknowledge or dispute relevant information.
F-10	Dual approval	Correction is applied only after borrower and authorized administrator approval.
F-11	Audit trail	Original values, proposals, decisions, reasons and timestamps remain traceable.
F-12	Notifications	Relevant users are notified about loans, returns, deadlines, faults and correction decisions.
F-13	Reports	Authorized roles filter reports and export permitted data.
F-14	Themes	Dark/light themes work consistently on web and mobile.
F-15	Responsive UI	Web adapts to small screens; Flutter uses mobile-native layouts.

Suggested build order

- 1. Confirm roles, permissions and school condition/marks rubric.
- 2. Finalize database schema and Express.js authentication/authorization.
- 3. Build React shared layout, theme toggle, login and role-based navigation.
- 4. Implement inventory and mandatory inspection screens.
- 5. Implement lending, returns, borrower acknowledgement and dual-approval corrections.
- 6. Add notifications, reports, audit history and Flutter mobile screens.
- 7. Test permissions, transaction integrity, accessibility and backup/restore.

End of mockup blueprint

Mockups and sample data are illustrative. Confirm final roles, approval policy, responsibility-mark rubric, due-date rules and school data classifications before implementation.